IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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9 November 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data Collection through API and Web Scraping
- Data Wrangling
- Exploratory Data Analysis with SQL and Data Visualization
- Interactive Visual Analytics with Folium
- Plotly Dash dashboard
- Machine Learning Prediction:
  - Trained Logistic Regression, SVM, Decision Tree, and KNN using GridSearchCV tuning

Summary of all results

- Exploratory Data Analysis result
- Interactive analytics in screenshots
- Predictive Analytics result

Introduction

Project background and context

SpaceX advertises Falcon 9 rocket launches on its website at a cost of $62 million, whereas competing providers charge upwards of $165 million per launch. A major portion of these cost savings stems from SpaceX's ability to reuse the rocket's first stage.

Predicting booster landings enables accurate launch cost estimation, giving competing aerospace firms the financial intelligence needed to bid effectively against SpaceX. This project builds an end-to-end machine learning pipeline to accurately forecast first-stage recovery outcomes.

Problems Statement

- Which operational factors most strongly dictate booster recovery?
- Can machine learning models accurately forecast first-stage landing outcomes from historical data?
- How can landing probabilities be used to model launch costs for competitive market bidding?

Section 1

Methodology



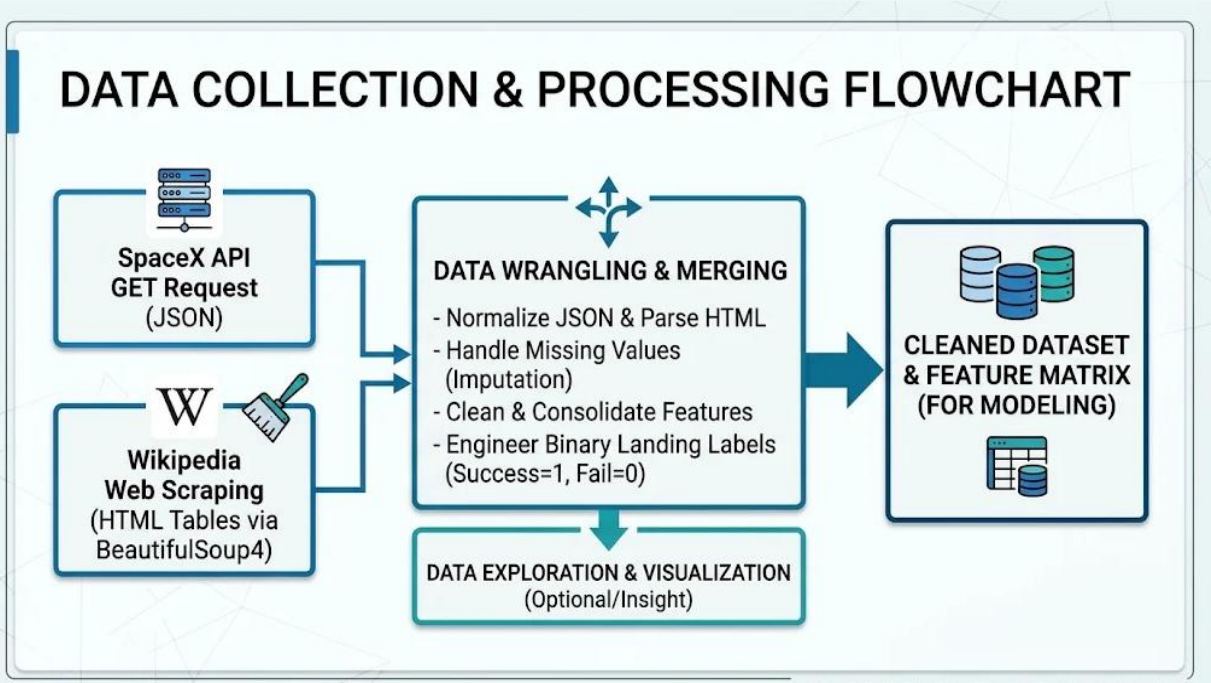
Methodology

Executive Summary

- Data collection methodology:
  - Data was collected using SpaceX API and web scraping from Wikipedia.
- Perform data wrangling
  - Handling missing values, encoding categorical variables, and engineering binary landing success labels for modelling.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Classification models were built, hyperparameter-tuned, and evaluated to predict landing success rates.

Data Collection

- **SpaceX API Extraction:**
  - Queried launch data via GET requests
  - Decoded JSON payloads, and converted them into Pandas DataFrames.
- **Wikipedia Scraping:**
  - Used BeautifulSoup to parse HTML tables from Wikipedia for historical Falcon 9 launch records.
- **Data Preprocessing:**
  - Cleaned the combined dataset by identifying and imputing missing values for future analysis.

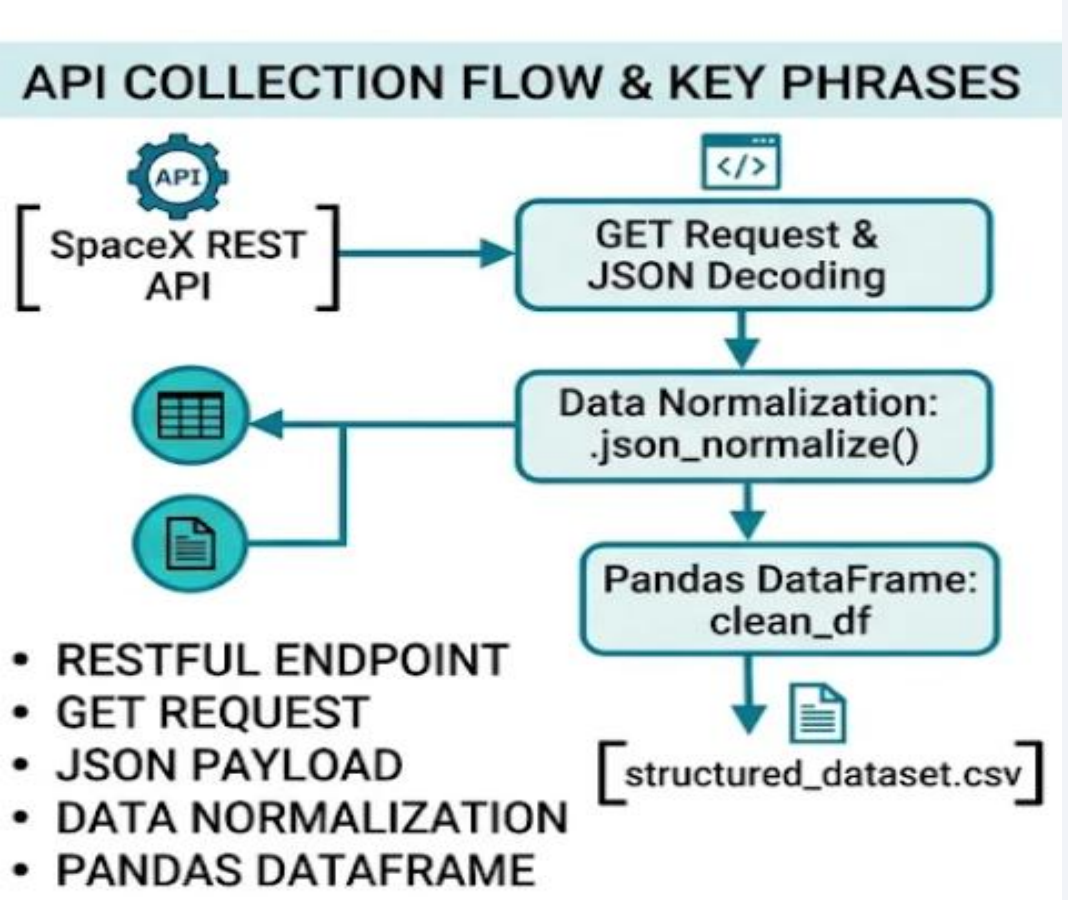


Data Collection – SpaceX API

Key Phrases & Process

- **API Endpoint Query:**
  - Executed HTTP GET requests to retrieve raw JSON launch records from the SpaceX API.
- **JSON Decoding:**
  - Processed response content using the .json() function call.
- **Data Normalization:**
  - Flattened nested JSON structures into a structured Pandas DataFrame using .json_normalize()

Reference: [GitHub Notebook: SpaceX API Collection]



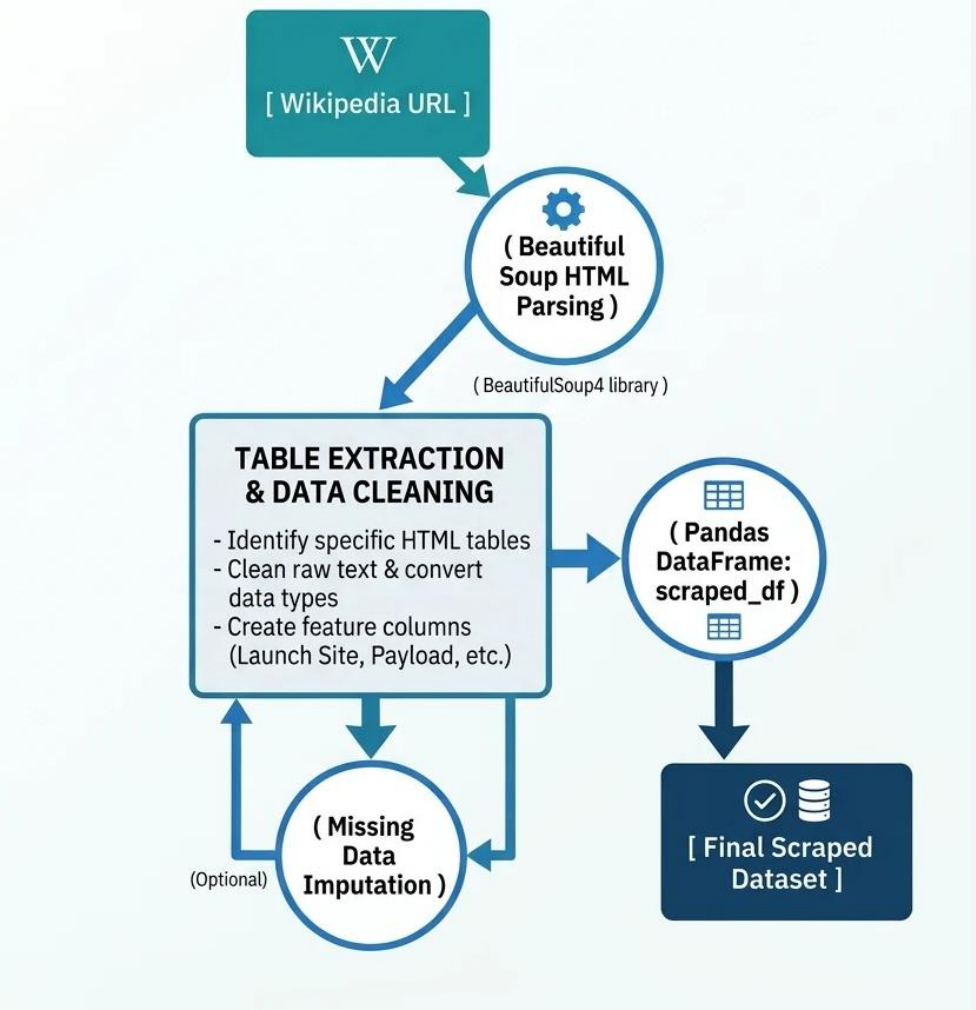
Data Collection - Scraping

Key Phrases & Process

- **Web Scraping Execution:** Extracted historical Falcon 9 launch records from Wikipedia using BeautifulSoup4.

- **HTML Table Parsing:** Structured and converted raw HTML table data into a clean Pandas DataFrame for downstream analysis.

Reference: [GitHub Notebook: SpaceX Web Scraping]



Data Wrangling

Exploratory Data Analysis

- Evaluated column data types, checked for missing attributes (e.g., LandingPad), and analyzed launch site and orbit distributions.

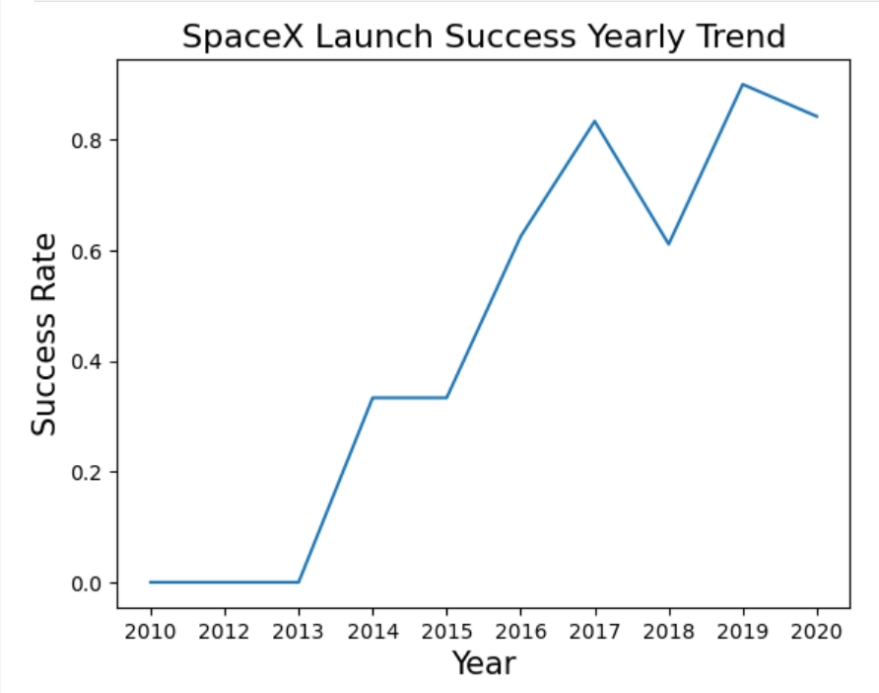
- Mapped specific landing outcomes (Outcome) into binary training labels (Class)
  - Successful landings are labeled as 1
  - Unsuccessful attempts are labeled as 0.

- Calculated the overall success rate and exported the cleaned, label-engineered dataset to CSV.

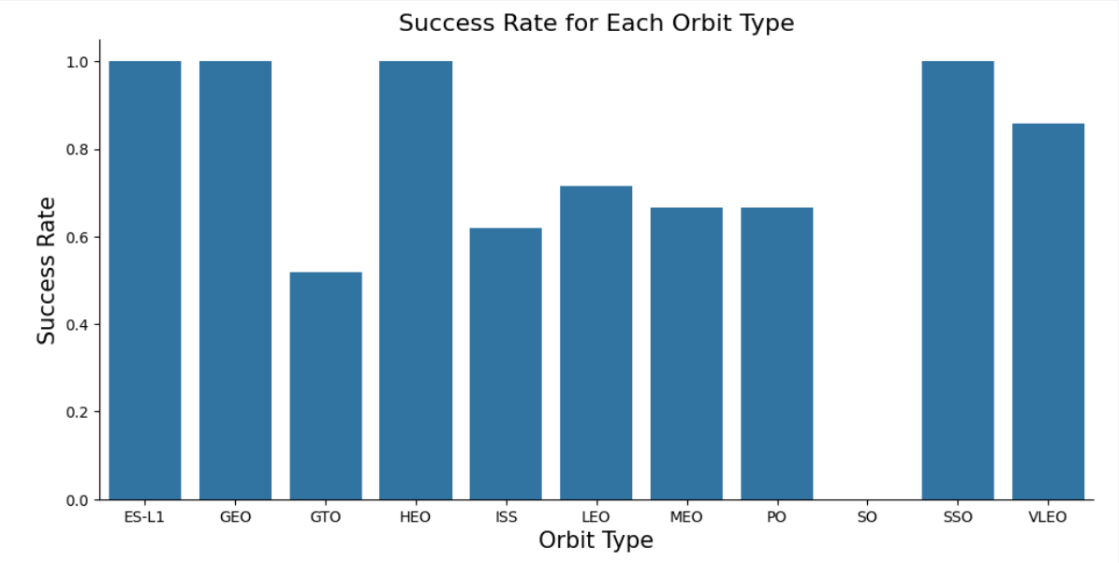
ite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

Reference: GitHub Notebook: SpaceX Data Wrangling

EDA with Data Visualization



- Tracks yearly landing success rates from 2013 to 2020 to evaluate overall operational improvement.



- Compares success rates across different orbital destinations to identify mission complexity and reliability.

Reference: [GitHub Notebook: SpaceX EDA with Data Visualization]

EDA with SQL

- Loaded the SpaceX dataset into a PostgreSQL database.

- **Launch Sites:**
  - Queried unique facilities (CCAFS, VAFB, KSC)
  - Filtered records using prefix patterns ('CCA')
  .
- **Payload Analysis:**
  - Computed for total payload for NASA and average payload for F9 v1.1.

- **Landing Milestones:**
  - Tracked the first ground pad success (2015-12-22)
  - Filtered drone ship successes within a 4k–6k kg payload range.

- **Aggregations & Grouping:**
  - Summarized outcome counts across all missions
  - Used subqueries to identify max payload boosters.

- **Date & Range Filtering:**
  - Extracted specific 2015 drone ship failures using string slicing
  - Ranked outcomes between 2010 and 2017.

Reference: GitHub Notebook: SpaceX EDA with SQL

Build an Interactive Map with Folium

- **Launch Facility & Outcome Mapping:**
  - Visualized all launch sites with 1 km boundary circles
  - Encoded mission outcomes into binary classes (1 for success, 0 for failure).

- **Reliability Evaluation:**
  - Deployed color-coded marker clusters (green for success, red for failure) to easily identify and compare success rates across different launch sites.

- **Proximity & Infrastructure Analysis:**
  - Calculated straight-line distances (PolyLine) to evaluate geographic relationships with coastlines, highways, railways, and urban safety buffers.

Reference: GitHub Notebook: SpaceX Build an Interactive Map with Folium

Build a Dashboard with Plotly Dash

- **Interactive Launch Site Dropdown (**dcc.Dropdown**):**
  - Added a selector component to filter mission data by specific launch sites.

- **Success Proportion Pie Chart (**dcc.Graph **&** px.pie**):**
  - Implemented a pie chart to compare total success distributions across sites or breakdown success versus failure counts for a selected launch location.

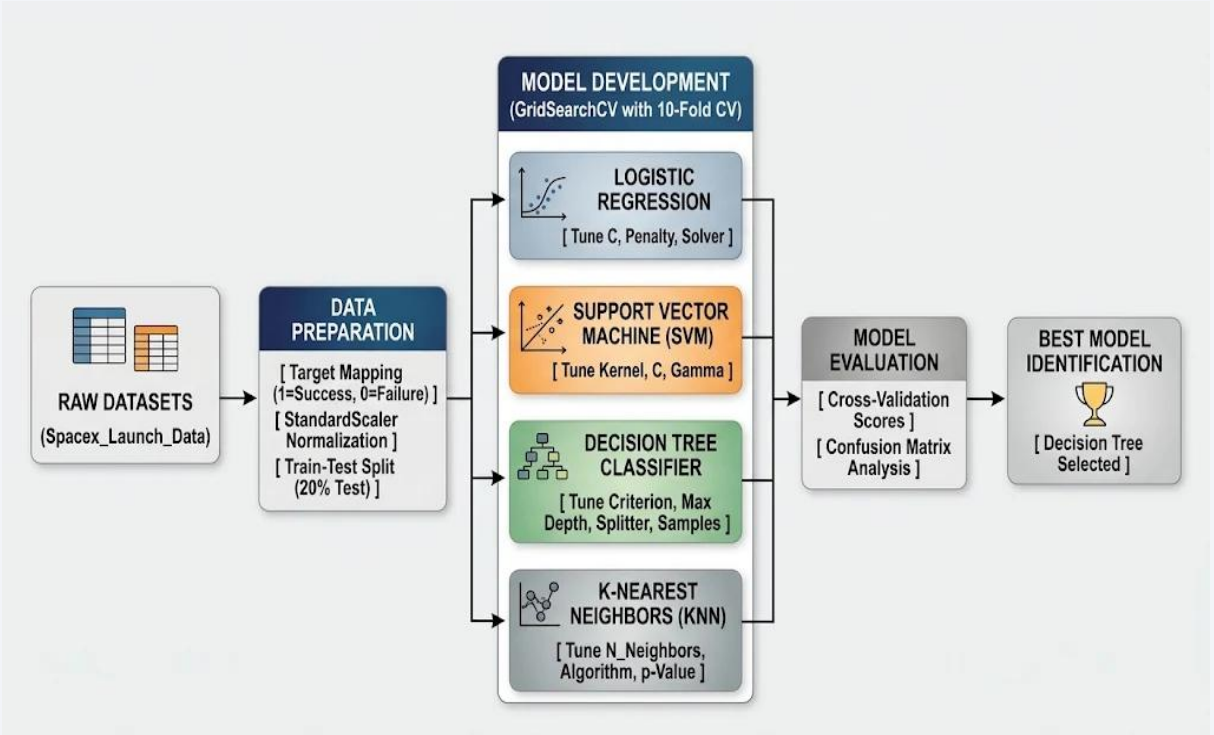
- **Payload Range Slider (**dcc.RangeSlider**):**
  - Added a numerical slider component to configure minimum and maximum payload weight boundaries.

- **Payload vs. Success Scatter Chart (**dcc.Graph**):**
  - Integrated an interactive correlation scatter plot linked to site selection and payload range inputs to evaluate the impact of payload mass and booster versions on launch outcomes.

Reference: GitHub Notebook: SpaceX Build a Dashboard with Plotly Dash

Predictive Analysis (Classification)

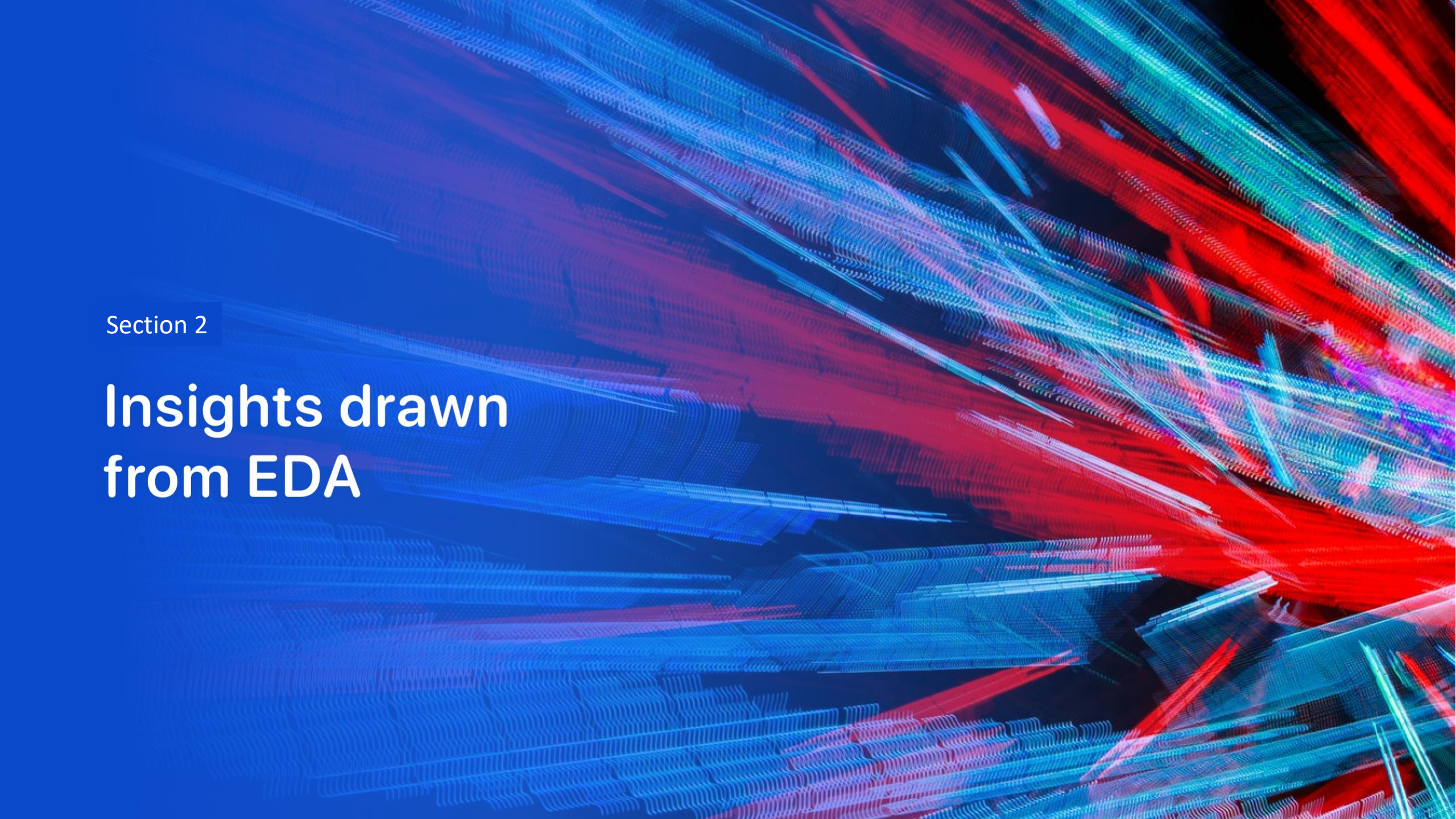
- **Data Preparation:**
  - Standardized feature columns and split data into training and test sets (test_size=0.2).
- **Model Training & Tuning:**
  - Trained Logistic Regression, SVM, Decision Tree, and KNN models using 10-fold cross-validation (GridSearchCV).
- **Performance Evaluation:**
  - Assessed test accuracies and confusion matrices to analyze classification errors.
- Identified the top-performing model based on cross-validation scores.



Reference: [GitHub Notebook: SpaceX Predictive Analysis (Classification)]

Results

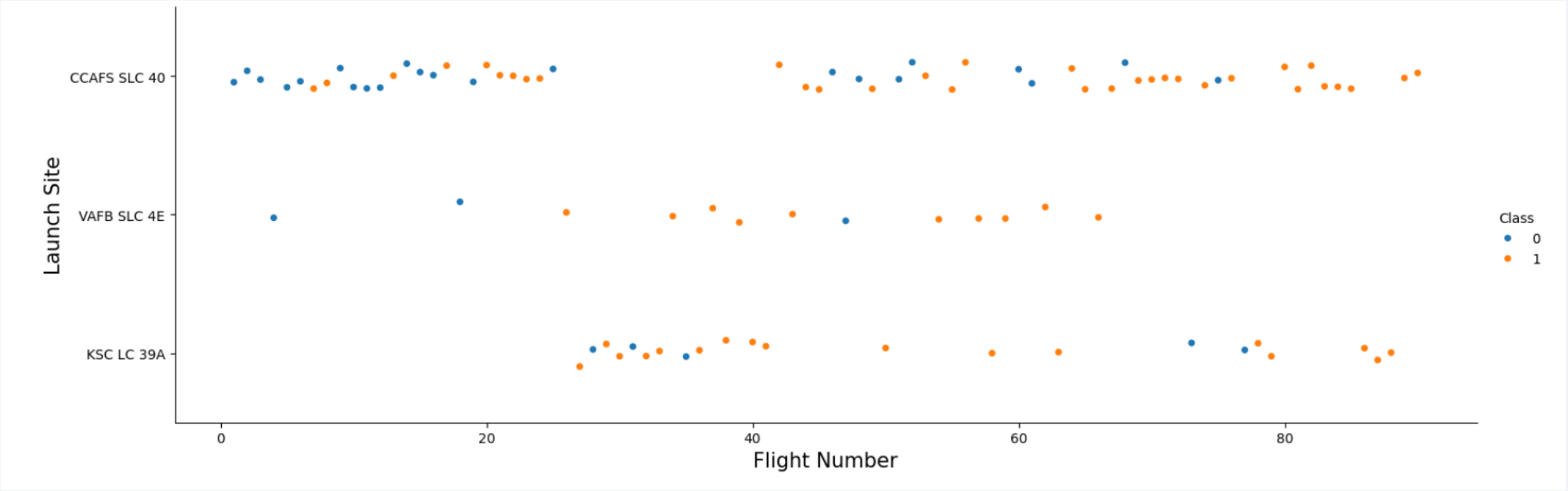
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results


Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

Scatter Plot of Flight Number vs. Launch Site

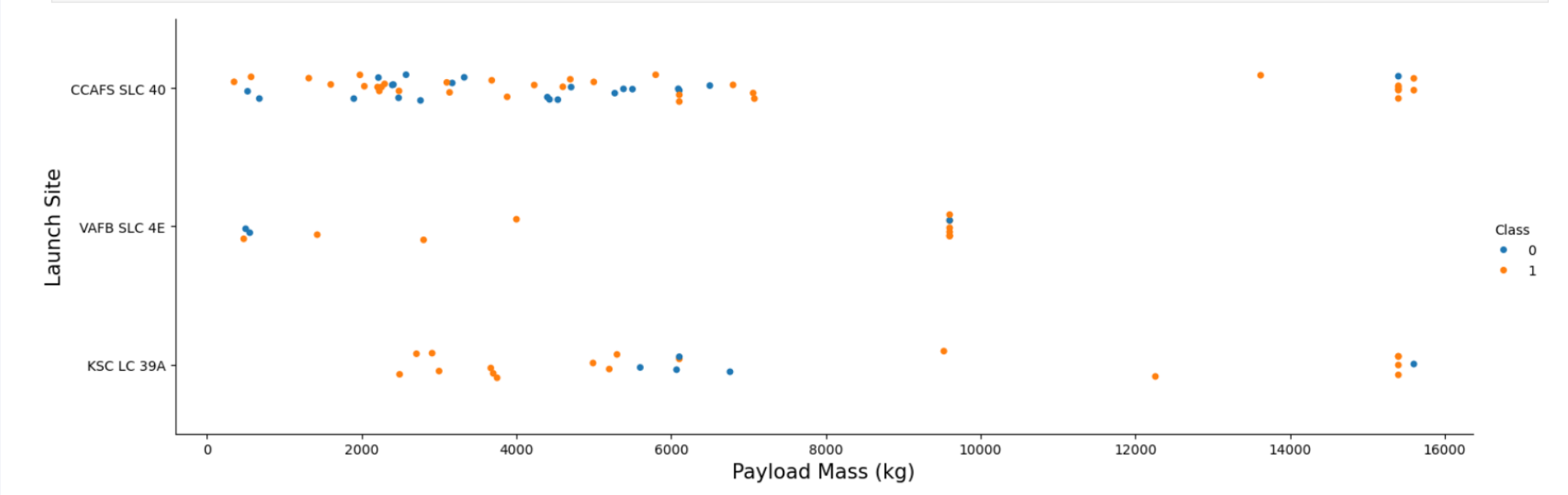
- The larger the flight amount at a launch site, the greater the success rate at a launch site.



Payload vs. Launch Site

Scatter Plot of Payload Mass vs. Launch Site

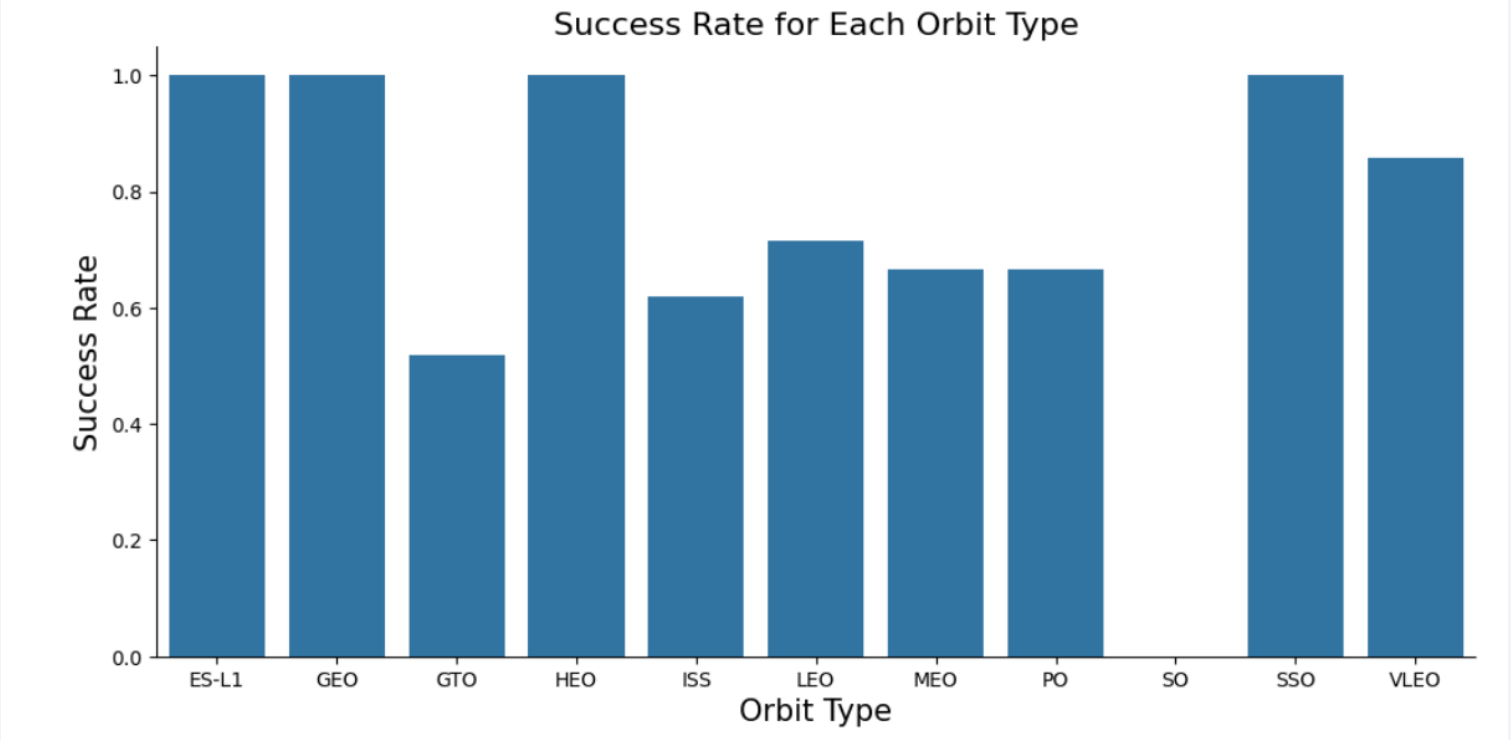
- Heavier payloads (>9,000 kg) show high success rates across multiple launch sites.
- CCAFS SLC 40 handles the broadest spectrum of payload masses, while KSC LC 39A frequently manages heavier missions.
- Lighter payloads (<5,000 kg) exhibit mixed success and failure outcomes, indicating that payload mass alone does not dictate landing reliability.



Success Rate vs. Orbit Type

Bar chart of Success Rate of each Orbit Type

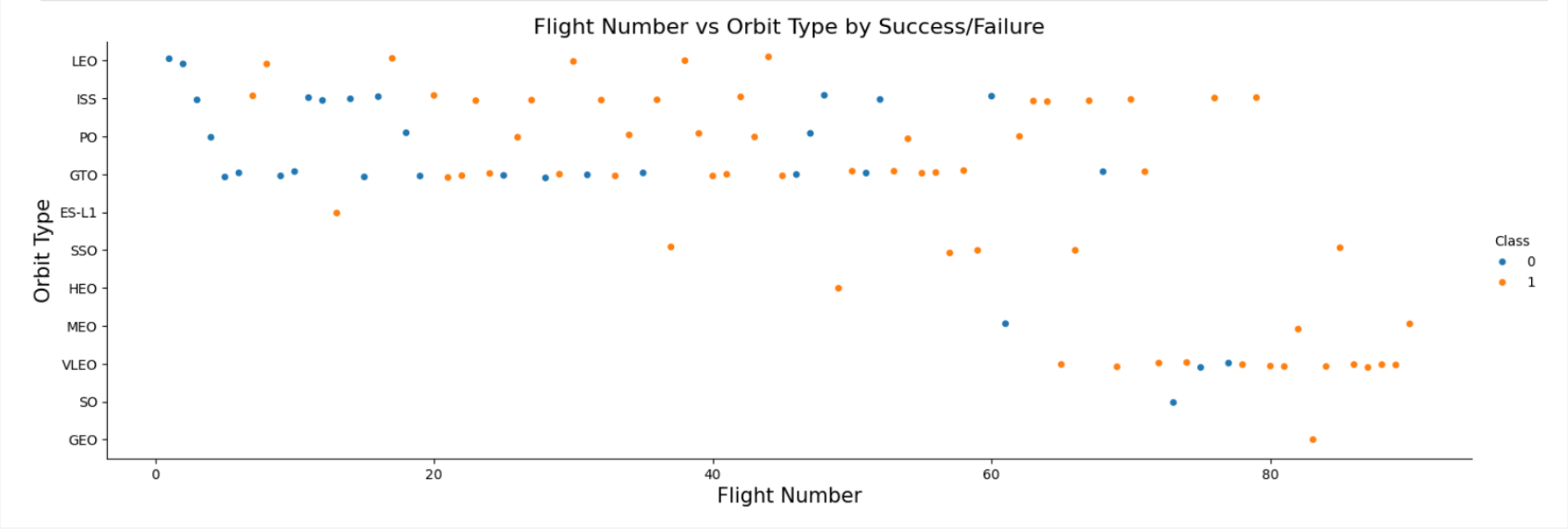
- ES-L1, GEO, HEO, SSO, VLEO had the most success rate.



Flight Number vs. Orbit Type

Scatter point of Flight Number vs. Orbit Type

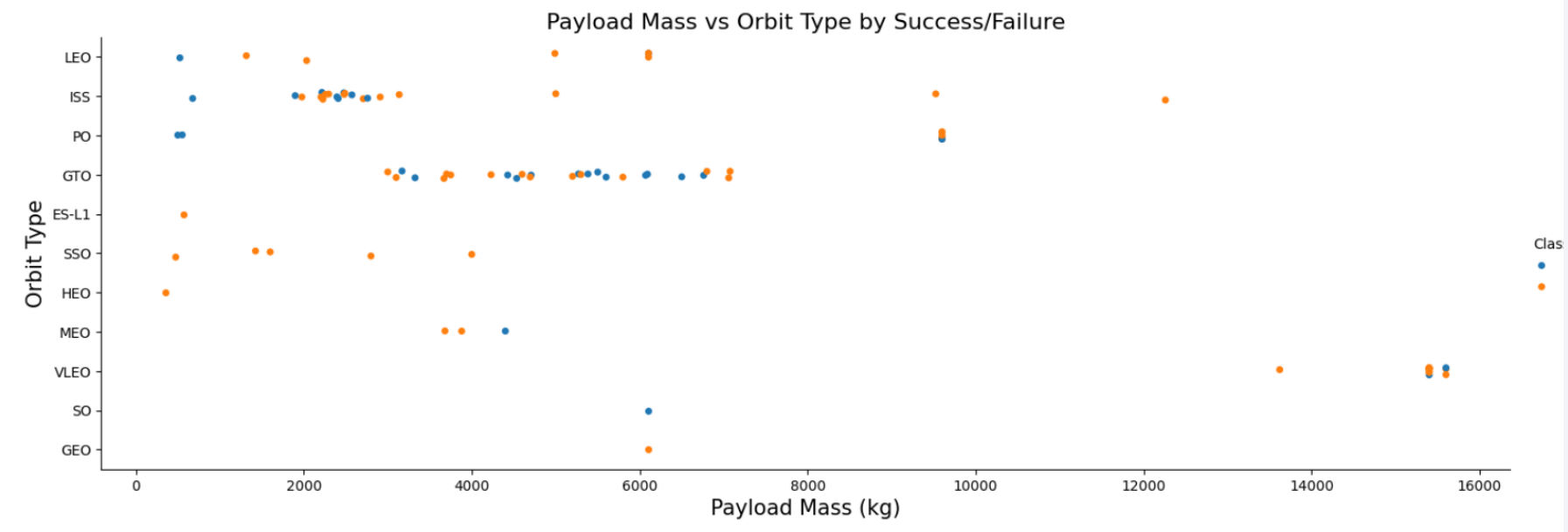
- Success rates improve as flight numbers increase (early flights show failures, while later flights transition mostly to successes).
- Success and failure outcomes are mixed evenly across flight numbers for GTO, indicating no strong correlation with mission experience.
- Higher flight numbers across various orbits (like VLEO and ISS) generally demonstrate improved landing reliability over time.



Payload vs. Orbit Type

Scatter point of Payload Mass vs. Orbit Type

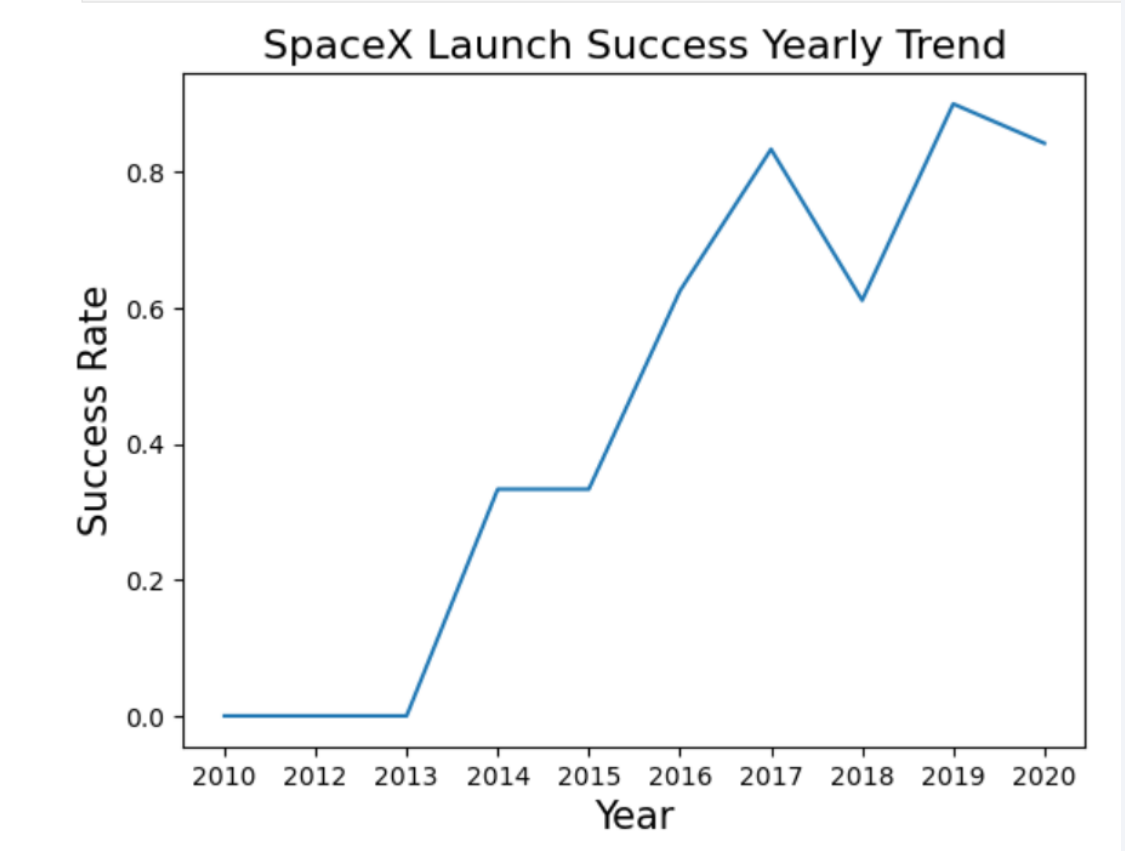
- **ES-L1, SSO, HEO, and GEO** show near 100% landing success across light-to-medium payload masses (<4,000 kg).
- **VLEO** and **ISS** with heavy payloads (>9,000 kg to ~15,000 kg) demonstrate consistently high landing success.
- **GTO** show a balanced mix of successful and failed landings, showing that payload mass alone does not guarantee landing success.



Launch Success Yearly Trend

Line chart of yearly average success rate

- **Year 2010–2013:** Landing success rate was **0%** as first-stage booster recovery was still experimental.
- **Positive Upward Trend:** Success rate increased sharply from **2013 to 2017**.
- **High Operational Reliability (2017–2020):** Maintained a high landing success rate consistently, reaching a peak in **2019**.



All Launch Site Names

- Executed SELECT DISTINCT Launch_Site via %sql magic in Jupyter Notebook.
- Extracted the 4 unique launch site names:
  - **CCAFS LC-40**
  - **VAFB SLC-4E**
  - **KSC LC-39A**
  - **CCAFS SLC-40**

In [11]:
```
%%sql
SELECT DISTINCT "Launch_Site"
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[11]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Executed an SQL query using LIKE 'CCA%' with LIMIT 5 to retrieve the first five launch records originating from CCAFS LC-40.

In [12]:
```
%%sql
SELECT * FROM SPACEXTABLE
    WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[12]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Used SUM() on PAYLOAD_MASS_KG_ for NASA missions.

- **Total payload mass carried by boosters launched by NASA: 48,213 kg**

In [13]:
```
%%sql
SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE
    WHERE "Customer" LIKE '%NASA (CRS)%';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

SUM("PAYLOAD_MASS__KG_")
48213

Average Payload Mass by F9 v1.1

- Used AVG(PAYLOAD_MASS_KG_) filtered by Booster_Version LIKE 'F9 v1.1%'.

- **Average payload carried by Falcon 9 v1.1 rockets.: 2,928.4 kg**

In [14]:
```
%%sql
SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE
    WHERE "Booster_Version" = 'F9 v1.1'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

AVG("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

- Used MIN(Date) filtered by Landing_Outcome = 'Success (ground pad)'.

- **First Ground Landing Date: 2015-12-22** (marking SpaceX's historic first successful booster touchdown on land at LZ-1).

In [15]:
```
%%sql
SELECT MIN("Date") FROM SPACEXTABLE
    WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Filtered Landing_Outcome = 'Success (drone ship)' where PAYLOAD_MASS__KG_ BETWEEN 4001 AND 5999.

- **Identified Boosters:**
  - **F9 FT B1022**
  - **F9 FT B1026**
  - **F9 FT B1021.2**
  - **F9 FT B1031.2**

- Full Thrust (FT) booster variants successfully executed ocean barge recoveries while lifting medium-heavy satellite payloads (GTO missions).

In [16]:
```
%%sql
SELECT "Booster_Version" FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS__KG_" > 4000
  AND "PAYLOAD_MASS__KG_" < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Used SELECT Landing_Outcome, COUNT(*) AS Total_Number ... GROUP BY Landing_Outcome
- **Success Outcomes (Total: 61):**
- **Failure Outcomes (Total: 10):**
- Successful recovery outcomes significantly outnumber failures (61 vs. 10), reflecting high overall landing reliability as technology matured.

In [20]:
```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS "Total_Number" FROM SPACEXTABLE
GROUP BY "Landing_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Landing_Outcome	Total_Number
Controlled (ocean)	5
Failure	3
Failure (drone ship)	5
Failure (parachute)	2
No attempt	21
No attempt	1
Precluded (drone ship)	1
Success	38
Success (drone ship)	14
Success (ground pad)	9
Uncontrolled (ocean)	2

Boosters Carried Maximum Payload

- Used a subquery WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE).

- **Boosters Identified with Maximum Payload Mass: Falcon 9 Block 5 (F9 B5)** boosters
  - B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7.

- Upgraded Block 5 variants frequently handle maximum payload capacities while maintaining reusability across multiple flights.

In [21]:
```
%%sql
SELECT "Booster_Version" FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (
    SELECT MAX("PAYLOAD_MASS__KG_")
    FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Out[21]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Extracted records using substr(Date, 1, 4) = '2015' and filtered for Landing_Outcome = 'Failure (drone ship)'.
- **Identified Failures (2015):**
  - **Month 01 (Jan):** Booster F9 v1.1 B1012 launched from CCAFS LC-40.
  - **Month 04 (Apr):** Booster F9 v1.1 B1015 launched from CCAFS LC-40.

In [22]:
```
%%sql
SELECT substr("Date", 6, 2) AS "Month", "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE
WHERE substr("Date", 1, 4) = '2015'
  AND "Landing_Outcome" = 'Failure (drone ship)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[22]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Used COUNT(Landing_Outcome) filtered by Date BETWEEN '2010-06-04' AND '2017-03-20', grouped and ordered by Total DESC.

- **Ranked Results (Descending):**
  - No attempt: 10
  - Success (drone ship): 5
  - Failure (drone ship): 5
  - Success (ground pad): 3
  - Controlled (ocean): 3
  - Uncontrolled (ocean): 2
  - Failure (parachute): 2
  - Precluded (drone ship): 1

In [23]:
```
%%sql
SELECT "Landing_Outcome", COUNT("Landing_Outcome") AS "Total" FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY "Total" DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[23]:

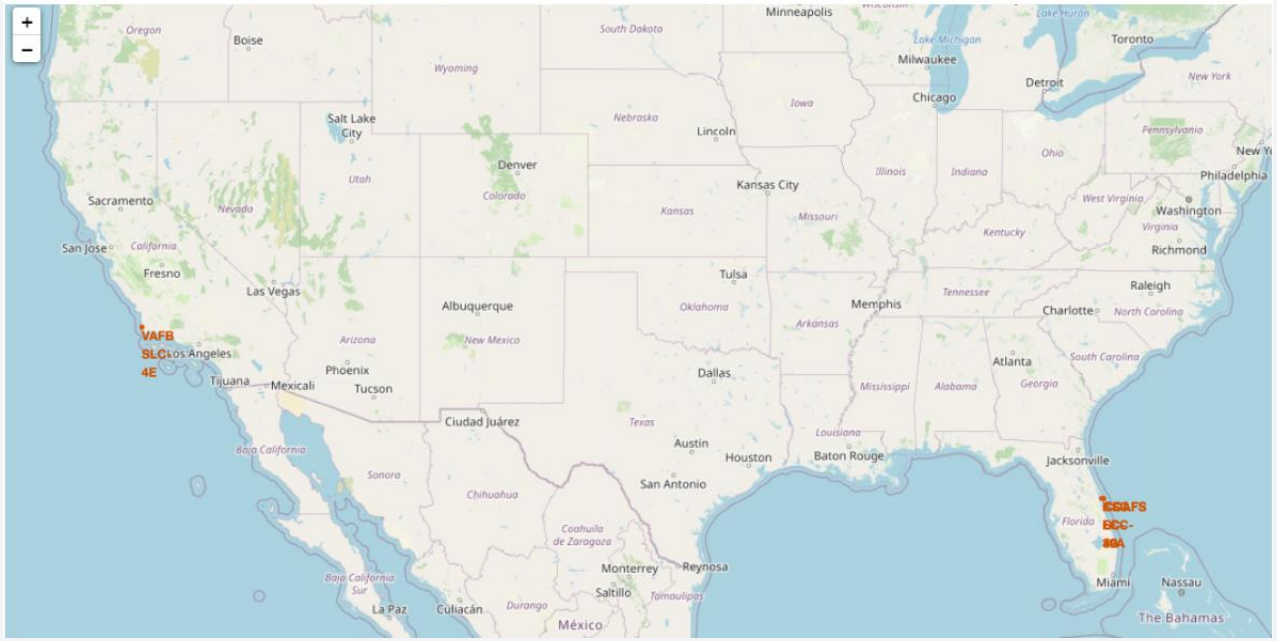
Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

Global Overview of SpaceX Launch Sites

- All launch sites are located directly along coastline borders (East Coast of Florida & West Coast of California).

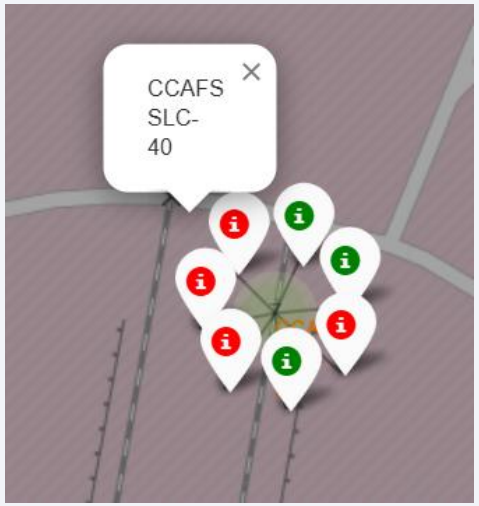
- **Identified Launch Facilities:**
  - **West Coast:** VAFB SLC-4E for polar orbit trajectories.
  - **East Coast:** CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A for equatorial and ISS missions.

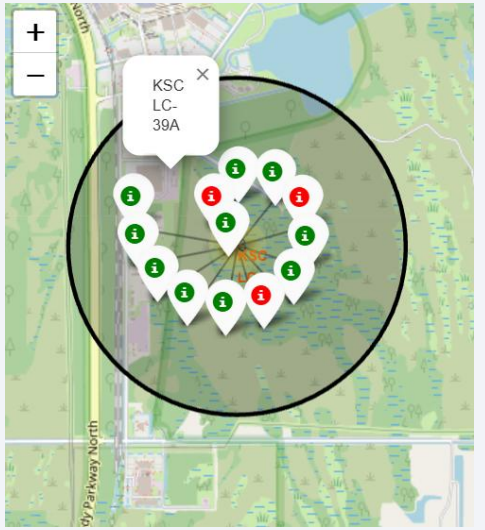


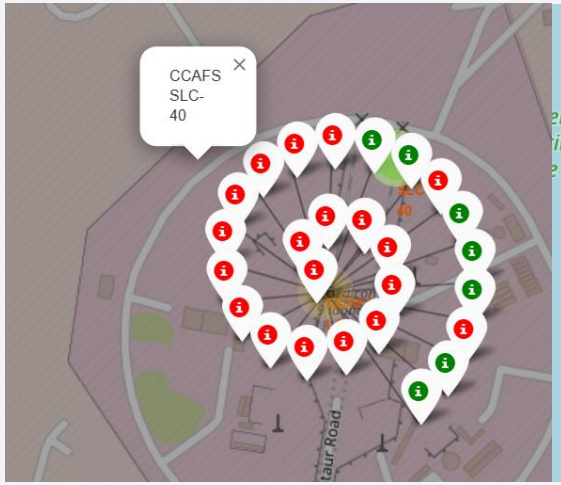
Site-Specific Launch Outcomes Breakdown (Folium Map)

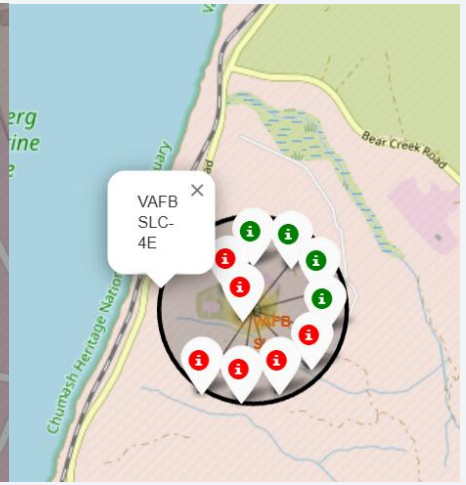
- **Color-Coded Marker Legend:**
  - **Green Icons (class = 1):** Successful first-stage booster landing.
  - **Red Icons (class = 0):** Failed booster landing or unrecovered mission.

- **Site-by-Site Findings:**
  - **KSC LC-39A:** Highest concentration of green success markers.
  - **VAFB SLC-4E:** Displays a near-even mix of green and red markers.
  - **CCAFS LC-40 / SLC-40:** Displays the highest volume of early red failure markers from initial testing phases, transitioning to higher green success ratios in later launches.

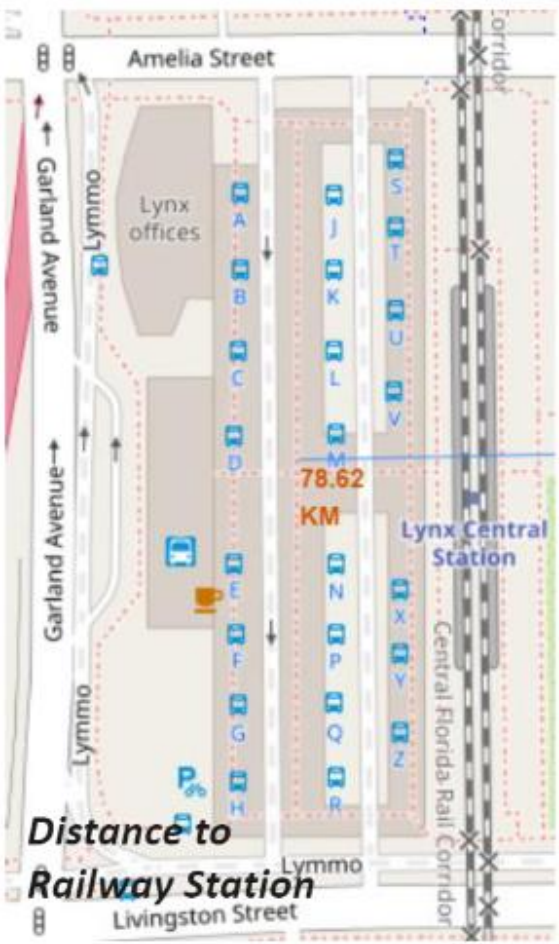






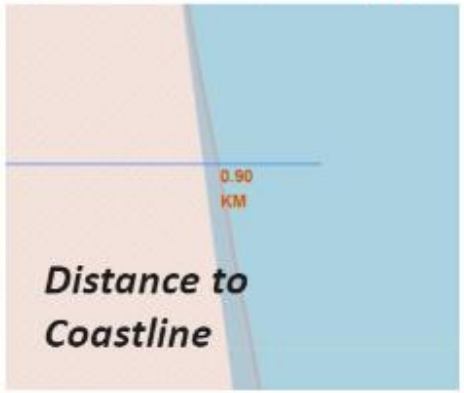


Launch Site Infrastructure & Proximity Analysis



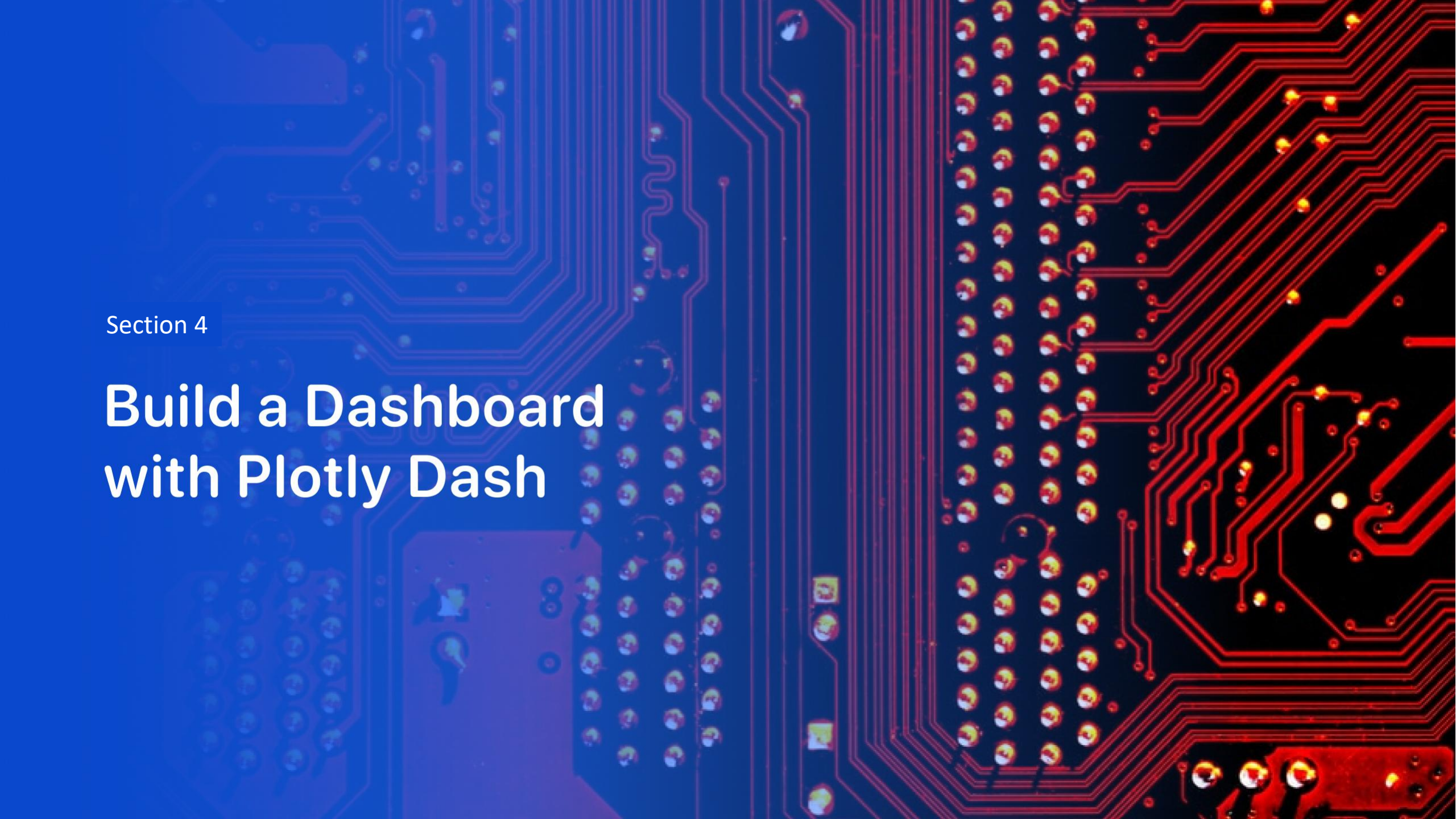








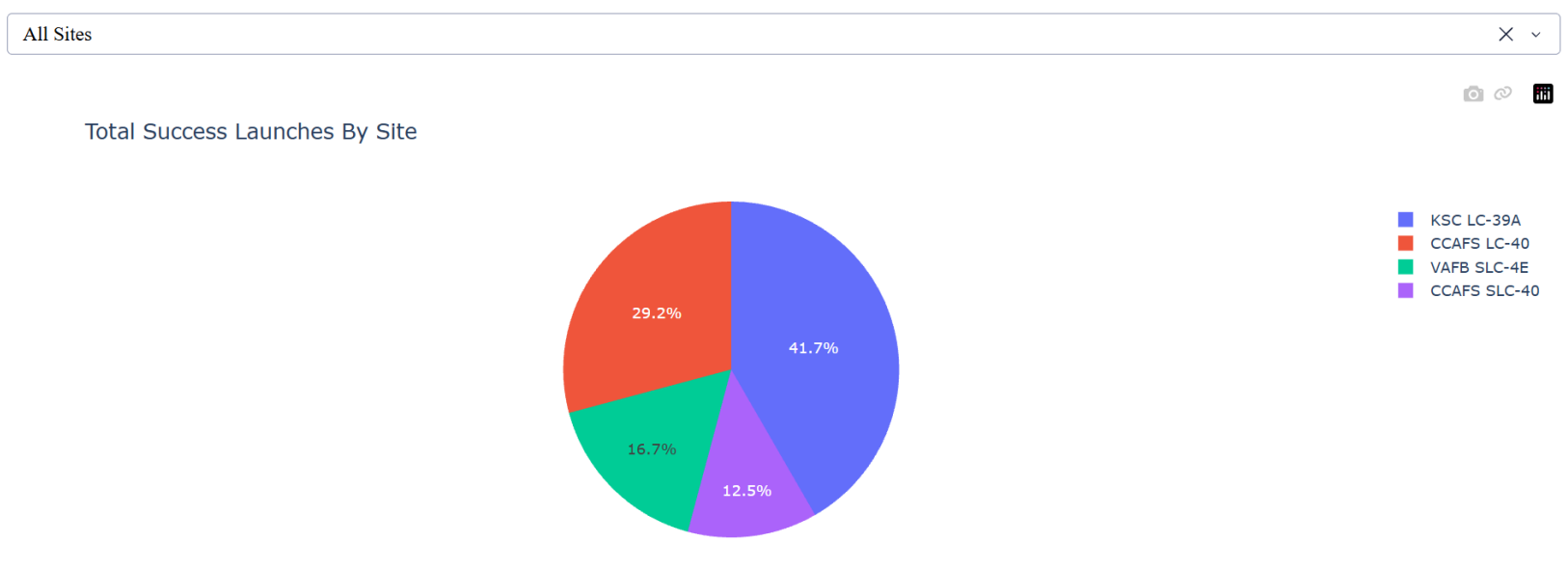
- Are launch sites in close proximity to railways? No
- Are launch sites in close proximity to highways? No
- Are launch sites in close proximity to coastline? Yes
- Do launch sites keep certain distance away from cities? Yes


Section 4
Build a Dashboard with Plotly Dash

Total Success Launches by Site (All Launch Sites)

Interactive Pie Chart: Visualizes the proportional contribution of successful launches across all four active SpaceX launch pads (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40).

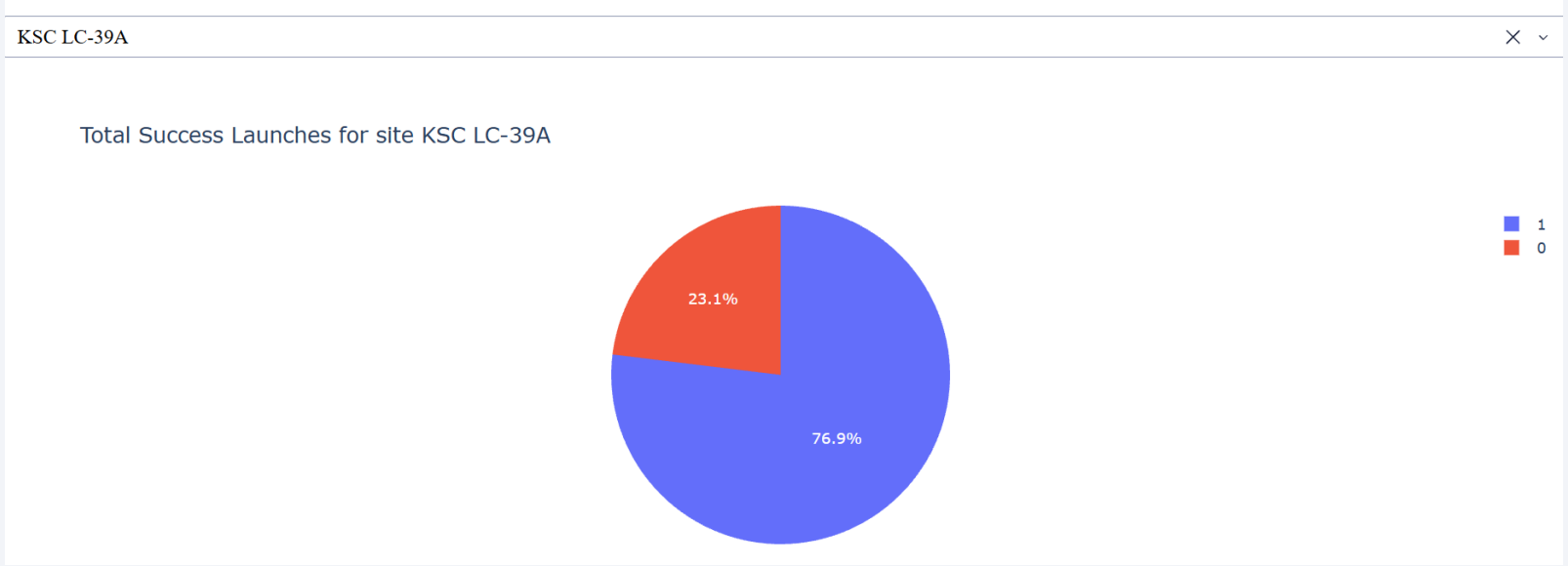
- **KSC LC-39A** accounts for the highest proportion of successful missions.
- **CCAFS LC-40** also handles a heavy launch volume, though its success ratio was impacted by early test-phase failures.



Launch Success vs. Failure Ratio for KSC LC-39A

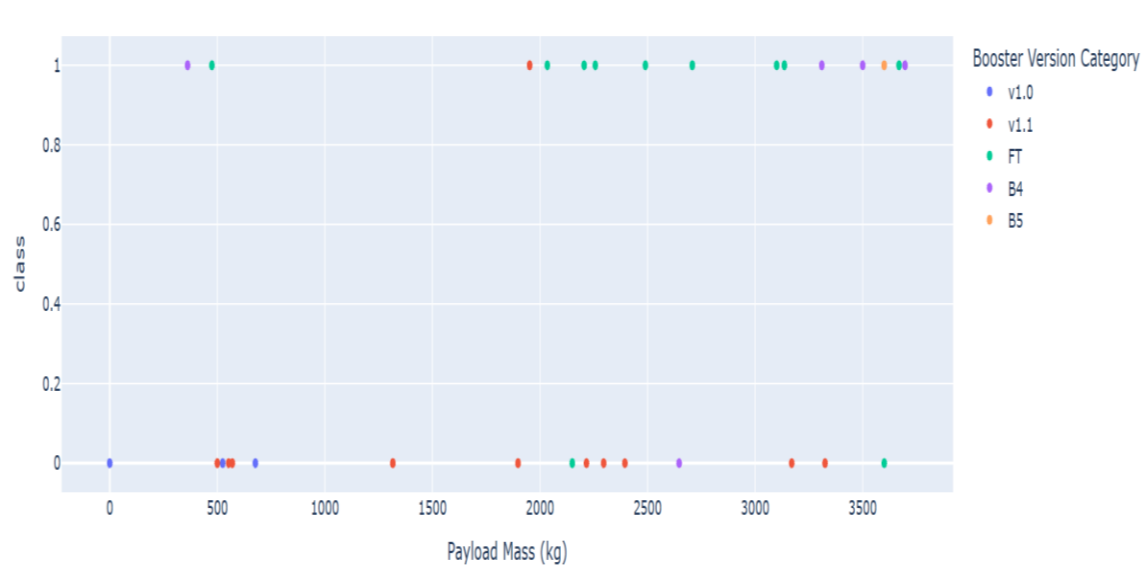
Pie chart breakdown of **Success (class 1)** vs **Failure (class 0)**.

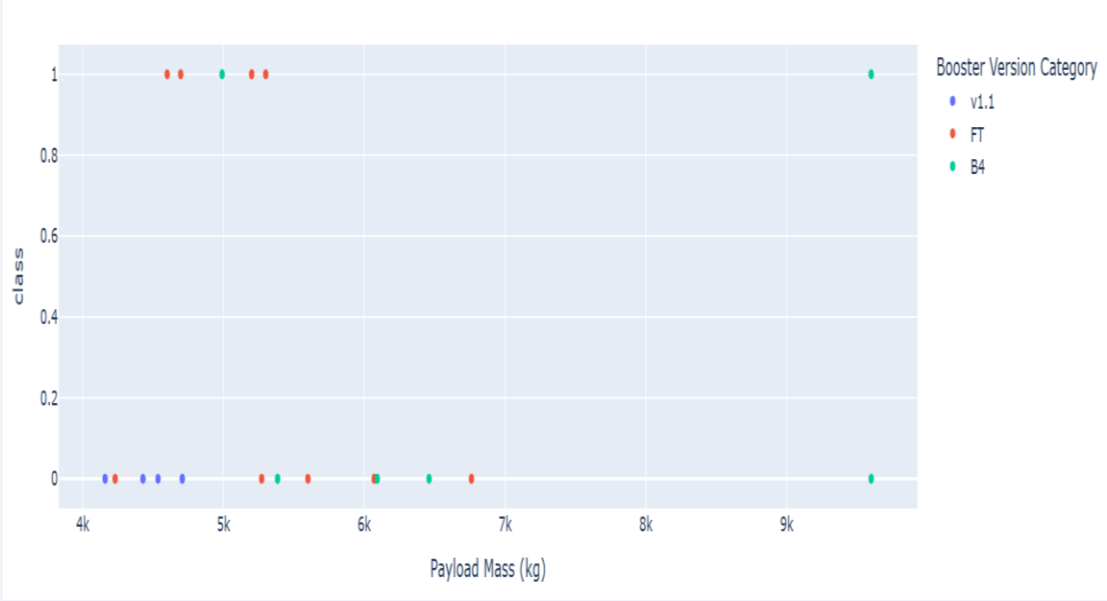
- **Highest Success Rate: KSC LC-39A** achieves the highest success ratio (~76.9%) of all sites.
- Reflects the reliability of Block 5 boosters utilized for high-priority ISS Dragon resupply and commercial payload missions.



Payload vs. Launch Outcome Correlation Analysis

- **Low-to-Medium Payloads ($0--4{,}000$ kg):** High success rates across early and mature booster versions (FT, B4).
- **Medium-Heavy Payloads ($4{,}000--6{,}000$ kg):** Mixed outcomes (successes and failures) depending on booster version and landing mechanism.
- **Heavy Payloads ($> 9{,}000$ kg):** Consistent high success rates driven by upgraded FT and B5 boosters.





Section 5

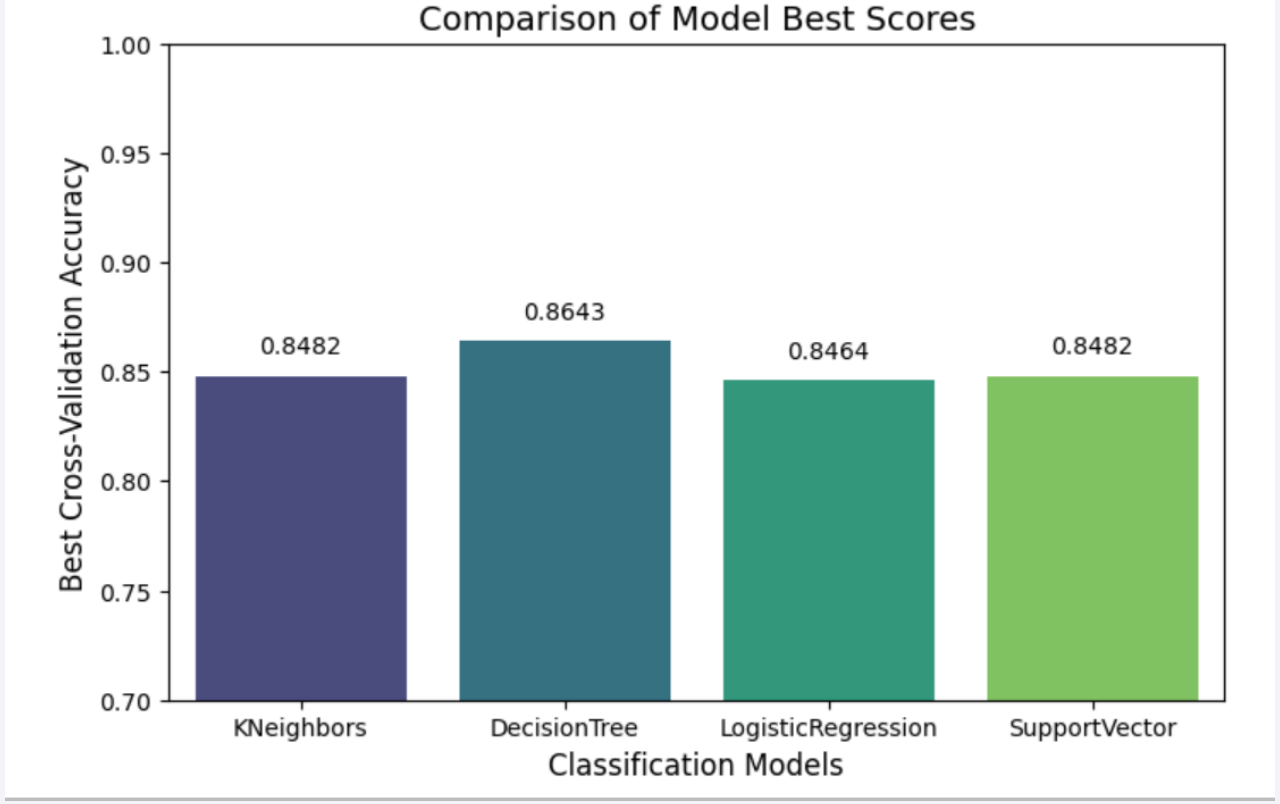
Predictive Analysis (Classification)



Classification Accuracy

- **Model Comparison & Visualization:**
  - Evaluated four classification algorithms using 10-fold cross-validation (GridSearchCV): **DecisionTree**, **KNeighbors**, **LogisticRegression**, and **SupportVector**.

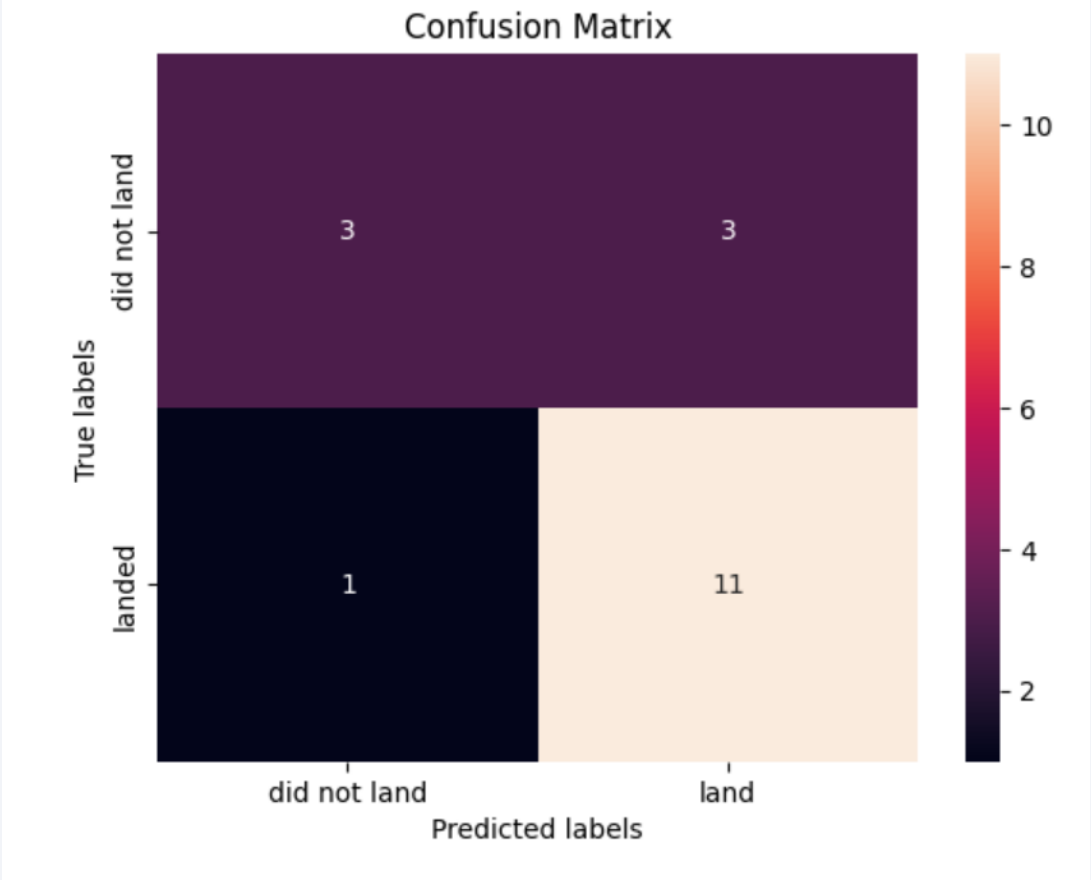
- **Decision Tree Classifier** achieved the highest cross-validation score of **86.43%**.



Confusion Matrix

The confusion matrix for the decision tree classifier demonstrates a strong ability to distinguish between the two launch outcomes.

However, the primary challenge lies in false positives—specifically, cases where unsuccessful first-stage landings were incorrectly predicted as successful.



Conclusions

- Historical data analysis identified key performance metrics, revealing KSC LC-39A as the launch site with the highest success ratio.
- Payload mass and upgraded booster versions (such as FT and B5) strongly correlate with landing success.
- Folium mapping confirmed that launch pads are strategically positioned near coastlines and required safety buffers.
- Rigorous model evaluation identified the Decision Tree Classifier as the best-performing model (~88.9% cross-validation accuracy).

Thank you!

